

Institutional Delta Sweeps [BOSWaves]

```
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// © BOSWaves
```

```
//@version=6
indicator("Institutional Delta Sweeps [BOSWaves]", overlay = true, max_labels_count = 500,
max_boxes_count = 500, max_lines_count = 500, calc_bars_count = 3000, max_bars_back =
5000)
```

// BOSWaves — Inputs

//

```
const string GRP_SWING = "Swing Detection"
swingLeft  = input.int(20, "Pivot Lookback Left", minval = 5, group = GRP_SWING, tooltip =
"Number of bars to the left used to identify a major swing high/low.")
swingRight = input.int(20, "Pivot Lookback Right", minval = 5, group = GRP_SWING, tooltip =
"Number of bars to the right required to confirm a swing point.")
cooldownBars = input.int(10, "Sweep Cooldown Bars", minval = 0, group = GRP_SWING, tooltip =
"Minimum bars required between detecting sweeps of the exact same zone to prevent alert spam.")
```

```
const string GRP_LIQ = "Liquidity Zones"
showLiqZones  = input.bool(true, "Show Liquidity Pools", group = GRP_LIQ, tooltip = "Toggles the
visibility of the liquidity blocks and lines.")
liqZoneWidth  = input.float(0.3, "Zone ATR Width", minval = 0.1, maxval = 1.0, step = 0.1, group =
GRP_LIQ, tooltip = "Defines the vertical depth of the liquidity block.")
liqLineWidth  = input.int(1, "Liquidity Line Thickness", minval = 1, maxval = 5, group = GRP_LIQ)
maxZoneAge    = input.int(200, "Max Zone Age (bars)", minval = 50, group = GRP_LIQ, tooltip =
"How many bars a liquidity pool persists before fading out.")
```

```
const string GRP_SWEEP = "Sweep Visuals & Zones"
showSweepZone    = input.bool(true, "Highlight Sweep Zone", group = GRP_SWEEP, tooltip =
"Draws a localized box highlighting the exact candle wick that swept the liquidity.")
showSweepData    = input.bool(true, "Show Sweep Labels",   group = GRP_SWEEP, tooltip =
```

```
"Displays a text label showing the exact percentage the liquidity zone was penetrated.")
showBuySellZones = input.bool(true, "Project Buy/Sell Zones After Sweep", group = GRP_SWEEP,
tooltip = "Draws explicitly labeled, forward-projecting action zones immediately following a sweep.")
zoneExtension    = input.int(50, "Buy/Sell Zone Projection (Bars)", minval=10, group=GRP_SWEEP,
tooltip = "How many bars into the future the Buy/Sell zone boxes are projected.")
zoneBorderWidth  = input.int(1, "Zone Border Thickness", minval = 1, maxval = 5, group =
GRP_SWEEP, tooltip = "Adjusts the thickness of the dashed borders on the projected Buy/Sell
zones.")
```

```
const string GRP_ZZ = "Structure"
showZigZag    = input.bool(true, "Show Market Structure", inline = "zz", group = GRP_ZZ, tooltip =
"Draws lines connecting the major swing highs and lows to visualize market structure.")
zzStyle       = input.string("Dotted", "", ["Dotted", "Dashed", "Solid"], inline = "zz", group =
GRP_ZZ)
zzWidth       = input.xint(1, "Thickness", minval = 1, maxval = 5, inline = "zz", group = GRP_ZZ,
tooltip = "Adjusts the thickness of the structure/zigzag lines.")
```

```
const string GRP_CLR = "Colors"
bullColor    = input×color(#089981, "Bullish / Buy Zone", inline = "clr", group = GRP_CLR, tooltip =
"Color used for support lines, uptrend structure, and Buy Zones.")
bearColor    = input×color(#F23645, "Bearish / Sell Zone", inline = "clr", group = GRP_CLR, tooltip =
"Color used for resistance lines, downtrend structure, and Sell Zones.")
sweepBullColor = input×color(#089981, "Sweep Bull Highlight", inline = "clr2", group = GRP_CLR,
tooltip = "Color of the flash/highlight when a bullish sweep occurs.")
sweepBearColor = input×color(#F23645, "Sweep Bear Highlight", inline = "clr2", group = GRP_CLR,
tooltip = "Color of the flash/highlight when a bearish sweep occurs.")
```

BOSWaves — Helpers

//

|

```
getLineStyle() =>
  switch zzStyle
    "Dotted" => line.style_dotted
    "Dashed" => line.style_dashed
    => line.style_solid
```

```
// ─────────────────────────────────── BOSWaves ─ Types
```

```
//
```

```
type LiquidityZone
```

```
    float price
```

```
    int  startIndex
```

```
    bool isHigh
```

```
    bool swept
```

```
    int  sweepIndex
```

```
    float sweepWick
```

```
    float sweepVol
```

```
    line levelLine
```

```
    box  poolBox
```

```
    box  poolBoxInner
```

```
    box  sweepHighlight
```

```
type SweepStats
```

```
    int  bullCount
```

```
    int  bearCount
```

```
    float avgBullDepth
```

```
    float avgBearDepth
```

```
type SwingPoint
```

```
    float price
```

```
    int  idx
```

```
// ─────────────────────────────────── BOSWaves ─ State
```

```
//
```

```
var zones    = array.new<LiquidityZone>()
```

```
var stats    = SweepStats.new(0, 0, 0.0, 0.0)
```

```
var int lastBullSweepBar = 0
var int lastBearSweepBar = 0
```

```
var highSwing = SwingPoint.new(na, na)
var lowSwing  = SwingPoint.new(na, na)
var isDownMove = false
```

```
atr = taxatr(200)
```

// BOSWaves — Pivot Detection

// _____

```
pHigh = ta.pivohigh(high, swingLeft, swingRight)
pLow  = ta.pivotlow(low, swingLeft, swingRight)
```

```
H = ta.highest(high, swingLeft)
L = ta.lowest(low, swingLeft)
```

```

if high == H
    isDownMove := true
if low == L
    isDownMove := false

```

```

if high[1] == H[1] and high < H
  highSwing.idx := bar_index[1]
  highSwing.price := high[1]

```

```

if low[1] == L[1] and low > L
  lowSwing.idx := bar_index[1]
  lowSwing.price := low[1]

```

// _____ BOSWaves — ZigZag Structure

```
if showZigZag and isDownMove != isDownMove[1]
  zzColor = isDownMove ? bearColor : bullColor
  line.new(highSwing.idx, highSwing.price, lowSwing.idx, lowSwing.price, color =
color×new(zzColor, 60), style = getLineStyle(), width = zzWidth)
```

```
// ┌─────────────────────────────────────────────────────────────────────────────────── BOSWaves ── Create Liquidity
Zones ────────────────────────────────────────────────────────────────────────────────────┘
```

```
//
```

```
if not na(pHigh) and showLiqZones
  pivotPrice = high[swingRight]
  pivotIdx   = bar_index[swingRight]
  zoneTop    = pivotPrice + (atr × liqZoneWidth)
  zoneMid     = pivotPrice + (atr × liqZoneWidth × 0.3)
```

```
zone = LiquidityZone.new()
zone.price    := pivotPrice
zone.startIdx := pivotIdx
zone.isHigh   := true
zone.swept    := false
```

```
zone.poolBox := box.new(pivotIdx, zoneTop, bar_index + 10, pivotPrice, border_color =
color×new(bearColor, 60), bgcolor = color.new(bearColor, 90))
zone.poolBoxInner := box.new(pivotIdx, zoneMid, bar_index + 10, pivotPrice, border_color = na,
bgcolor = color.new(bearColor, 85))
zone.levelLine := line.new(pivotIdx, pivotPrice, bar_index + 10, pivotPrice, color =
color×new(bearColor, 40), style = line×style_solid, width = liqLineWidth)
```

```
zones.push(zone)
```

```
label.new(pivotIdx, pivotPrice, "◆",
  style    = label×style_none,
  textcolor = color.new(bearColor, 30),
  size     = sizextiny)
```

```

if not na(pLow) and showLiqZones
    pivotPrice = low[swingRight]
    pivotIdx   = bar_index[swingRight]
    zoneBot    = pivotPrice - (atr × liqZoneWidth)
    zoneMid    = pivotPrice - (atr × liqZoneWidth × 0.3)

    zone = LiquidityZone.new()
    zone.price      := pivotPrice
    zone.startIdx  := pivotIdx
    zone.isHigh    := false
    zone.swept     := false

    zone.poolBox := box.new(pivotIdx, pivotPrice, bar_index + 10, zoneBot, border_color =
color×new(bullColor, 60), bgcolor = color.new(bullColor, 90))
    zone.poolBoxInner := box.new(pivotIdx, pivotPrice, bar_index + 10, zoneMid, border_color = na,
bgcolor = color.new(bullColor, 85))
    zone.levelLine := line.new(pivotIdx, pivotPrice, bar_index + 10, pivotPrice, color =
color×new(bullColor, 40), style = linexstyle_solid, width = liqLineWidth)

    zones.push(zone)

    label.new(pivotIdx, pivotPrice, "◆",
        style    = label×style_none,
        textcolor = color.new(bullColor, 30),
        size     = sizextiny)

// ┌────────────────────────────────────────────────────────────────────────────────── BOSWaves — Zone Management
& Sweep Detection ───────────────────────────────────────────────────────────────────────────────────┐
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if zones.size() > 0
    for i = zones×size() - 1 to 0

        zone = zones×get(i)
        age  = bar_index - zone.startIdx

```

```
if zone.swept
    continue
```

```
if age > maxZoneAge
    box.delete(zone.poolBox)
    box.delete(zone.poolBoxInner)
    line.delete(zone.levelLine)
    zones.remove(i)
    continue
```

```
// ——— Comet Fade Math ———
```

```
fadePct    = math×min(age / maxZoneAge, 1.0)
cometCurve  = math×pow(fadePct, 3)
bgTransp    = int(90.0 + (10.0 × cometCurve))
innerTransp = int(85.0 + (15.0 × cometCurve))
borderTransp = int(60.0 + (40.0 × cometCurve))
lineTransp  = int(40.0 + (60.0 × cometCurve))
```

```
zoneColor = zonexisHigh ? bearColor : bullColor
```

```
box.set_right(zone.poolBox, bar_index + 5)
box.set_right(zone.poolBoxInner, bar_index + 5)
line.set_x2(zone.levelLine, bar_index + 5)
```

```
box.set_bgcolor(zone.poolBox, color.new(zoneColor, bgTransp))
box.set_bgcolor(zone.poolBoxInner, color.new(zoneColor, innerTransp))
box.set_border_color(zone.poolBox, color.new(zoneColor, borderTransp))
line.set_color(zone.levelLine, color.new(zoneColor, lineTransp))
```

```
// Bearish Sweep
```

```
if zone.isHigh and high > zone.price and close < zone.price and open < zone.price
    if bar_index >= lastBearSweepBar + cooldownBars
        zone.swept    := true
        zone.sweepIndex := bar_index
        zone.sweepWick  := high
        zone.sweepVol   := volume
        lastBearSweepBar := bar_index
```

```
box.set_right(zone.poolBox, bar_index)
box.set_right(zone.poolBoxInner, bar_index)
line.set_x2(zone.levelLine, bar_index)
```

```
if showSweepZone
```

```
    zone.sweepHighlight := box.new(
        bar_index - 1, high, bar_index + 1, zone.price,
        border_color = color.new(sweepBearColor, 40),
        border_width = zoneBorderWidth,
        bgcolor      = color.new(sweepBearColor, 85))
```

```
if showSweepData
```

```
    penetration = ((high - zone.price) / zone.price) × 100
    label.new(bar_index, high,
        "SWEEP " + str.tostring(penetration, "#.##") + "%",
        style    = label.style_label_down,
        color    = color.new(sweepBearColor, 30),
        textcolor = color.white,
        size     = size.small)
```

```
if showBuySellZones
```

```
    zoneTop = high
    zoneBot = math.max(open, close)
    box.new(left = bar_index, top = zoneTop, right = bar_index + zoneExtension, bottom =
zoneBot,
        border_color = color.new(bearColor, 40), border_width = zoneBorderWidth,
border_style = line.style_dashed,
        bgcolor = color.new(bearColor, 90),
        text = "SELL ZONE", text_color = color.new(bearColor, 40),
        text_halign = text.align_right, text_valign = text.align_top)
```

```
stats.bearCount += 1
```

```
penetration = ((high - zone.price) / zone.price) × 100
```

```
stats.avgBearDepth := ((stats.avgBearDepth * (stats.bearCount - 1)) + penetration) /
```

```
stats.bearCount
```

```
// Bullish Sweep
```



```

if not zone.isHigh and low < zone.price and close > zone.price and open > zone.price
    if bar_index >= lastBullSweepBar + cooldownBars
        zone.swept      := true
        zone.sweepIndex := bar_index
        zone.sweepWick  := low
        zone.sweepVol   := volume
        lastBullSweepBar := bar_index

box.set_right(zone.poolBox, bar_index)
box.set_right(zone.poolBoxInner, bar_index)
line.set_x2(zone.levelLine, bar_index)

if showSweepZone
    zone.sweepHighlight := box.new(
        bar_index - 1, zone.price, bar_index + 1, low,
        border_color = color.new(sweepBullColor, 40),
        border_width = zoneBorderWidth,
        bgcolor      = color.new(sweepBullColor, 85))

if showSweepData
    penetration = ((zone.price - low) / zone.price) × 100
    label.new(bar_index, low,
        "SWEEP " + str.tostring(penetration, "#.##") + "%",
        style    = label.style_label_up,
        color    = color.new(sweepBullColor, 30),
        textcolor = color.white,
        size     = size.small)

if showBuySellZones
    zoneTop = math.min(open, close)
    zoneBot = low
    box.new(left = bar_index, top = zoneTop, right = bar_index + zoneExtension, bottom =
zoneBot,
        border_color = color.new(bullColor, 40), border_width = zoneBorderWidth,
border_style = line.style_dashed,
        bgcolor = color.new(bullColor, 90),
        text = "BUY ZONE", text_color = color.new(bullColor, 40),
        text_halign = text.align_right, text_valign = text.align_bottom)

```

```

stats.bullCount += 1
penetration = ((zonexprice - low) / zonexprice) × 100
stats.avgBullDepth := ((stats.avgBullDepth * (stats.bullCount - 1)) + penetration) /
stats.bullCount

// ┌────────────────────────────────────────── BOSWaves — Alerts
// └──────────────────────────────────────────┘
// ┌──────────────────────────────────────────────────────────────────────────────────
// └──────────────────────────────────────────────────────────────────────────────────┘

bullSweepSignal = false
bearSweepSignal = false

if zones.size() > 0
    for z in zones
        if z.swept and z.sweepIndex == bar_index
            if z.isHigh
                bearSweepSignal := true
            else
                bullSweepSignal := true

alertcondition(bullSweepSignal, title = "Bullish Sweep", message = "{{ticker}} Bullish Liquidity
Sweep at {{close}}")
alertcondition(bearSweepSignal, title = "Bearish Sweep", message = "{{ticker}} Bearish Liquidity
Sweep at {{close}}")

```